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Calcium Phosphate Nanoparticles for Transcutaneous Vaccine Delivery

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The main objective of this study was to investigate the potential of calcium phosphate (CAP) nanoparticles for transcutaneous vaccine delivery. CAP nanoparticles were prepared by nanoprecipitation method followed by sequential adsorption of sugars and ovalbumin. Nanoparticles were characterized using dynamic light scattering, XRD, ATR-FTIR, and microscopy methods. *In-vitro* release of ovalbumin from nanoparticles was studied in phosphate buffer (pH 7.4). *In-vivo* immunization studies were carried out in Balb/C mice. The size and zeta potential of ovalbumin-sugar adsorbed CAP nanoparticles was 350 ± 22.5 nm and −12.93 ± 1.02 mV respectively. Around 60% ovalbumin was released from nanoparticles within 24 hrs. To test the feasibility for transcutaneous vaccine delivery, the nanoparticles were applied in mice after removing the stratum corneum by tape-stripping. In the positive control group, the nanoparticles were administered by intradermal injection. Ovalbumin-sugar coated CAP nanoparticles showed significantly higher antibody titers (Total IgG and IgG1) compared to ovalbumin alone. IgG2a antibodies were only seen with intradermal injection. Both topical and intradermal treatment groups showed splenocyte proliferation when re-stimulated with ovalbumin. The results from this study demonstrate the potential of using CAP nanoparticles for transcutaneous vaccine delivery.

Keywords: CAP Nanoparticles, Adjuvant, Transcutaneous Vaccination, Tape Stripping.

1. INTRODUCTION

Skin is a potential site for vaccination since the epidermal and dermal layers of the skin are rich in antigen-presenting cells including Langerhans cells (LCs) and dermal dendritic cells (DCs).[1] However, the outermost layer of skin, stratum corneum (SC), acts as the major barrier and limits the penetration of larger molecules (>500 Da) including antigens and adjuvants into the skin.[2] Glenn et al.,[3] demonstrated the delivery of *Escherichia coli* heat-labile enterotoxin (LT) via simple hydration of the skin using wet gauze. Later, it was shown that pretreatment with emery paper or tape stripping produced higher anti-LT antibodies compared to pretreatment with pumice stone or hydration alone.[4] Tape stripping is a simple, inexpensive and mild procedure to remove the epidermal barrier.[5] Also, it activates LCs and keratinocytes resulting in higher immune response.[6] Controlled SC disruption has been used in humans for vaccine delivery.[7] Alternatively several penetration enhancement approaches have been explored for transcutaneous immunization (TCI).[8] These include the use of electrical methods, ultrasound and intradermal vaccine delivery through jet injectors and microneedle arrays.[1] Vaccine delivery via simple topical application of patches can be self-administered, improve patient compliance, and provide an efficient and cost effective method for mass vaccination.[9] Co-administration of protein antigens along with vaccine adjuvants such as cholera toxin (CT) and LT enhances the antigen specific antibody responses through skin.[3,10,11] Although CT and LT are very potent biological adjuvants, they also have safety concerns, thus necessitating the need for alternative adjuvants.[12]

Currently, aluminum based mineral salts are the only approved adjuvants for human use in the United States. Alum has many drawbacks including IgE mediated allergic responses, occasional induction of local reactions such as erythema or granulomatous inflammation, variability in production of alum precipitated toxoids/vaccines, and

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stability issues during freeze drying.[13–16] Moreover, alum mainly induces humoral immune responses (Th2 type, IgG1 response) and very limited cellular immune response (Th1 type, IgG2a response).[16] Therefore, it has limited value in vaccination against viral infections or cancer which require cellular immunity.[13]

Calcium phosphate (CAP) is a normal constituent of the human body, present in bones and teeth. It is well tolerated and biodegradable.[17] CAP ceramics are widely used as bone substitutes.[18] CAP as a vaccine adjuvant has been approved in many European countries.[19] CAP adsorbed vaccines prolonged the immune stimulation with higher levels of IgG and lower IgE antibodies compared to alum adjuvanted vaccines.[14,20] Studies have shown significantly enhanced immune response with sugar coated CAP nanoparticles.[21–23] Sugars (especially disaccharides) stabilize the proteins during desiccation and prevent dehydration induced denaturation.[24] In addition, the sugar can serve as immunopotentiator, as it can bind to the mannose receptors on the surface of antigen presenting cells (APCs).[25] Although CAP nanoparticles have been used for vaccine delivery through subcutaneous, intramuscular, intraperitoneal injection and mucosal routes, they are yet to be explored for TCI. The convenience of skin application coupled to the potent adjuvant activity of CAP nanoparticles makes it a promising vaccine delivery approach. To this end the main objective of this study was to explore the feasibility of using CAP nanoparticles for TCI. CAP nanoparticles were coated with different sugars ovalbumin. To investigate if CAP nanoparticles can be used for TCI, the formulation was applied on tape-stripped skin in mice and the results were compared against intradermal injection.

2. MATERIALS AND METHODS

2.1. Materials

Calcium chloride, monobasic sodium phosphate, cellobiose, sucrose, trehalose, ovalbumin, aluminum hydroxide gel and all other chemicals were purchased from Sigma Aldrich (St. Louis, MO). Goat anti-mouse IgG/IgG1/IgG2a-HRP conjugate were purchased from Bethyl laboratories (Montgomery, TX).

2.2. Preparation of CAP Nanoparticles

CAP nanoparticles were prepared by nanoprecipitation method.[26–27] Briefly, 0.25 M monobasic sodium phosphate (pH 7.2) was added drop wise to 0.75 M calcium chloride under probe sonication (Sonics Vibracell, Newtown, CT), for 5 min at 4 °C. The dispersion was bath sonicated (Branson 1510, Danbury, CT) for 90 minutes followed by filtration through a nitrocellulose filter (pore size 0.22 μm; Sigma Aldrich, St. Louis, MO) to remove the larger particles. The resultant filtrate was lyophilized (VirTis, Gardiner, NY) and stored at 4 °C.

2.3. Preparation of Sugar and Ovalbumin Coated Nanoparticles

Sugar solution was added to lyophilized CAP particles (50 mg) and mechanically stirred for 1 hr followed by filtration through a nitrocellulose filter (pore size 0.22 μm). The filtrate was centrifuged at 5000 rpm for 10 min at 4 °C. The pellet was lyophilized. Ovalbumin solution in phosphate buffer (pH 7.4) was added to the lyophilized sugar coated CAP particles (50 mg) and stirred for 1 hr followed by filtration. The filtrate was lyophilized and stored at 4 °C.

2.4. Optimization of Sugar Adsorption

Three different disaccharides namely cellobiose, trehalose and sucrose were used at five different concentrations (0.5, 1, 2, 4 and 8 mg/ml). Sugar solution was added to CAP nanoparticles and stirred for 1 hr. The samples were removed and centrifuged at 5000 rpm for 10 min at 4 °C. The supernatant was used to measure the adsorption efficiency of sugars using anthrone-sulfuric acid assay.[28] Briefly, 40 μl of standard or sample was taken in a 96 well plate and kept at 4 °C for 15 min. Freshly prepared anthrone sulfuric acid reagent (2 mg/ml) was added to each well. The plate was kept at 92 °C for 3 min. Then the plate was kept for 5 min at room temperature followed by 15 min at 45 °C. Absorbance was read at 630 nm using a UV-Visible spectrophotometer (Spectra Max M2, Molecular Devices, Sunnyvale, CA). The concentration of sugars was calculated using the standard curve for respective sugars (concentration range $= 0.015$–0.5 mg/ml, $r^2 = 0.97$–0.99).

2.5. Optimization of Ovalbumin Adsorption

In order to optimize the adsorption of ovalbumin, five different concentrations (0.05, 0.1, 0.25, 0.5 and 1 mg/ml) of ovalbumin solution were added to sugar coated CAP nanoparticles and stirred for 1 hr at 4 °C. At the end of the study, the samples were removed, centrifuged at 10000 rpm for 10 min at 4 °C. The supernatant was used to measure the adsorption efficiency of ovalbumin using bicinchoninic acid (BCA) reagent and the absorbance was read at 565 nm in a UV-visible spectrophotometer (Spectra Max M2, Molecular Devices, Sunnyvale, CA). The concentration of ovalbumin was calculated using the standard curve (concentration range $= 0.02$–1.25 mg/ml, $r^2 = 0.99$).

Ovalbumin loading was calculated by the difference between the total amount of ovalbumin added and the unadsorbed amount in the supernatant. Loading efficiency was calculated by the following formula:

$$\text{Loading Efficiency} = [(\text{Total ovalbumin added} - \text{ovalbumin in supernatant}) / \text{Total ovalbumin added}] \times 100$$

2.6. Characterization of Nanoparticles

2.6.1. Dynamic Light Scattering (DLS)

Nanoparticles were characterized for size, polydispersity index and zeta potential using DLS (Nicomp 360 ZLS, Santa Barbara, CA). Particles were dispersed in deionized water at a concentration of 1 mg/ml for measuring the particle size. Zeta potential was measured in 10 mM HEPES buffer (pH 7.4).

2.6.2. X-Ray Diffraction (XRD)

The phase of CAP nanoparticles was characterized using powder XRD (Rigaku Ultima IV multipurpose XRD, Woodlands, Texas). The CAP nanoparticles were exposed to CuK radiation (40 kV × 44 mA). The angular range was 10–70° 2θ, and the counts were measured for 1 sec at each step. The instrument was operated in the step scan mode in increments of 0.03° 2θ.

2.6.3. Atomic Force Microscopy (AFM)

Microscopic images were obtained for CAP-cellobiose-ovalbumin nanoparticles using Nano-R2™AFM (Pacific nanotechnology, Santa Clara, CA). A drop of nanoparticles dispersed in phosphate buffer (500 μg/ml) was placed on a glass slide and allowed to dry overnight. The images were taken in tapping mode. The average particle size of fifty nanoparticles was determined using nanorule software.

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2.6.4. Attenuated Total Reflectance-Fourier Transform Infrared Spectroscopy (ATR-FTIR)

Nicolet 380™ ATR-FTIR spectrophotometer (Thermo electron Corporation, Madison, WI) was used for characterization of lyophilized nanoparticles (CAP, CAP-cellobiose, and CAP-cellobiose-ovalbumin). The IR spectrum was recorded on Zn selenide crystal at 4 cm^{-1} resolution and each spectrum was an average of 64 scans.

2.7. *In-Vitro* Release Studies

Ovalbumin coated nanoparticles (10 mg) were dispersed in phosphate buffer (pH 7.4) in eppendorf tubes. The tubes were incubated in a shaker water bath (Cole Parmer, Vernon Hills, IL) at 37 °C and 40 rpm. At respective time intervals (30 min, 1 hr, 2 hr, 4 hr, 8 hr, 16 hr, 24 hr, 48 hr) sample tubes were centrifuged at 10000 rpm for 10 min at 4 °C. The supernatant (100 μl) was removed and replaced with equal volume of fresh buffer. The ovalbumin concentration in the supernatant was analyzed using BCA reagent. The concentration of ovalbumin was calculated using the standard curve (concentration range = 0.02–1.25 mg/ml, $r^2 = 0.99$).

2.8. *In-Vivo* Immunization Studies

All *in-vivo* experiments were performed after approval from the Institutional Animal Care and use committee (IACUC) at South Dakota State University. Female Balb/C mice were purchased from Charles River laboratories (Wilmington, MA). Mice (5–6 weeks) were randomly divided into five groups with four mice in each group. Intradermal injection was carried out with the following formulations: (i) ovalbumin, (ii) alum-ovalbumin, (iii) CAP-cellobiose-ovalbumin nanoparticles. The amount of ovalbumin was kept constant at 20 μg in these treatment groups. Alum-ovalbumin (positive control group) was prepared by dissolving appropriate amount of ovalbumin in phosphate buffer (pH 7.4) and subsequent mixing with aluminium hydroxide gel followed by sonication for 5 min. The other two groups of mice were immunized topically (after tape stripping) with ovalbumin and CAP-cellobiose-ovalbumin nanoparticles. The amount of ovalbumin in these two groups was kept constant at 100 μg.

For topical immunization studies the mice abdominal skin was shaved using a hair clipper (Golden A5, Oster, Niles, IL) a day before the treatment. On the day of treatment, mice were anesthetized using isoflurane (VetEquip, Inc. Pleasanton, California). Tape stripping was performed using Scotch book tape (845, 3M, St. Paul, MN) by sequential removal of stratum corneum. To confirm the removal of stratum corneum Trans Epidermal Water Loss (TEWL) was measured after each tape strip using Vapometer (Delfin Technologies, Kuopio, Finland). The formulation was applied for 4 hrs. At the end of treatment, the formulation was removed and the mouse skin was wiped with a kim wipe. For all the treatment groups, the immunization protocol included a primary dose and a booster dose (5th week after the primary dose). Blood samples were collected from the retro orbital plexus at 0, 3rd and 5th week after primary and booster doses respectively. Samples were centrifuged at 4000 rpm for 30 min and the serum was stored at −80 °C until analyzed.

2.9. Analysis of Antibody Titers

Enzyme Linked Immunosorbent Assay (ELISA) was used to determine anti-ovalbumin antibodies (Total IgG, IgG1 and IgG2a) in the serum samples. Briefly, micro titer plates were coated with ovalbumin (1 μg/ well) in carbonate bicarbonate buffer (pH 9.6) and incubated overnight at 4 °C. The following day, wells were washed three times with wash solution (50 mM Tris, 0.14 M NaCl, 0.05% Tween 20, pH 8). Then the wells were incubated with 200 μl blocking solution (50 mM Tris, 0.14 M NaCl, 1% BSA, pH 8) for half an hour followed by washing three times with the wash solution. Serum Samples were serially diluted with sample diluent (50 mM Tris, 0.14 M NaCl, 1% BSA and 0.05% Tween 20, pH 8). The diluted samples (100 μl) were incubated in the plate for an hour followed by five times washing with the wash solution. HRP conjugate (1:6000, 100 μl) was incubated for an hour followed by 5 times washing. Finally the plates were incubated with 3, 3', 5, 5'-Tetramethylbenzidine substrate solution for 5 min. The reaction was stopped using 2 M H_2SO_4 and plate was read at 450 nm in the plate reader.

2.10. Splenocyte Proliferation Assay

To investigate the ability of memory T cells to recognize the antigen upon re-stimulation, *in-vitro* splenocyte proliferation assay was conducted. The mice were sacrificed at 5th week after the booster dose. Spleen was removed under sterile conditions and single cell suspension was prepared in RPMI 1640 medium supplemented with 10% heat inactivated fetal calf serum (FCS), glutamine, sodium pyruvate, 2 mercaptoethanol, penicillin and streptomycin.[29] The suspension was centrifuged at 2500 rpm at 25 °C for 5 min and the supernatant was discarded. NH_4Cl (0.1 M) was added to lyse the erythrocytes. The cells were again centrifuged after addition of RPMI 1640 medium. The pellet was redispersed in media and the cells were counted using cellometer (Nexcelom Bioscience, Lawrence, MA). Splenocytes were seeded in a 96 well plate at 10^6 cells/well cell density and treated with RPMI, ovalbumin (100 μg/ml) or positive control concanavalin A mitogen (Con A) (2.5 μg/ml) for 72 hrs. Con A is known to specifically stimulate the activation of T cell lymphocytes.[30–31] At the end of the treatment, the medium was removed and the cells were incubated with 50 μl of 3-(4, 5-Dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT) solution (0.5 mg/ml) for 4 hrs. After 4 hrs, 150 μl DMSO was added. The plate was read at 540 nm in a UV-Visible spectrophotometer. Stimulation index (SI) was calculated by dividing the absorbance value of cells treated with Con A or ovalbumin with that of RPMI treated cells.

2.11. Statistical Analysis

The statistical difference among different groups was performed using one-way ANOVA (Instat, Graph Pad Software Inc., La Jolla, CA) at a p value of <0.01 unless stated otherwise.

3. RESULTS AND DISCUSSION

3.1. Preparation and Characterization of Sugar-Ovalbumin Coated CAP Nanoparticles

CAP nanoparticles were prepared by the reaction between calcium chloride and monobasic sodium phosphate, under sonication.[27] The size of CAP particles varies with the method of preparation, time of stirring, and duration and frequency of sonication.[32] The particle size plays an important role in modulating the immune response via interaction with antigen presenting cells.[33] The reaction conditions including time of probe sonication (5 min), bath sonication (90 min), and temperature (4 °C) were optimized (data not shown) to prepare nanoparticles of defined size. Table I shows the average size, polydipsersity index and zeta potential of nanoparticles. The CAP nanoparticles were 40 ± 5 nm in size with 4 ± 0.5 mV zeta potential. The positive zeta potential of CAP nanoparticles is due to the calcium/phosphate ratio (3:1) used in this study and is consistent with literature reports.[22, 24, 34]

CAP is known to exist in various polymorphic forms including hydroxyapatite, brushite, tricalcium phosphate and amorphous.[35] Most types of calcium phosphates are insoluble/partially soluble in water, but soluble in acidic pH.[36] Formation of a particular polymorph of CAP depends on the reaction conditions including temperature, pH, concentration and ratio of inorganic cation to anion.[35, 37] Hydroxyapatite and tricalcium phosphate are formed at alkaline pH, while brushite is mainly formed at acidic pH.[35] In the present study, particles were prepared at pH 7.4. The phase of CAP nanoparticles was characterized using powder XRD (Fig. 1). The characteristic intense peaks at 25°, 31–32°, and 50–55° 2θ indicate that the CAP is predominantly in the hydroxyapatite form.[22, 25] FTIR spectrum (Fig. 2) also showed peaks for phosphate (949 cm^{-1}, 614 cm^{-1}, 493 cm^{-1}, 1105 cm^{-1}) and hydroxyl (3300–3500 cm^{-1}) groups characteristic of the hydroxyapatite phase.[22, 37]

The CAP nanoparticles were coated with polyhydroxyl disaccharides followed by lyophilization. The adsorption of sugar increased the size of the nanoparticles from 40 ± 5 nm to 75 ± 7.35 nm and decreased the zeta potential from 4 ± 0.5 mV to 0.61 ± 0.01 mV. This reduction in zeta potential is attributed to the negatively charged hydroxyl groups in sugars.[22] The broadening of OH stretching peaks at 3360 cm^{-1} in the FTIR spectra confirmed the adsorption of sugars on CAP nanoparticles. The disaccharides act as a cryoprotectant and preserve the conformation of proteins.[38] Lyophilization helps to promote the irreversible adsorption of sugars onto the CAP surface.[38] The binding of sugars to calcium phosphate core surface reduces the interfacial tension and lowers the free energy of the system, thereby making the system more stable.[25, 38] Based on the literature reports, three disaccharides including cellobiose,

Table I. Characteristics of particles measured using dynamic light scattering (DLS).

Formulations	Size (nm)	Polydispersity index	Zeta potential (mV)
CAP nanoparticles	40 ± 5	0.6 ± 0.05	4 ± 0.5
CAP-cellobiose nanoparticles	75 ± 7.35	0.39 ± 0.08	0.61 ± 0.01
CAP-cellobiose-ovalbumin nanoparticles	350 ± 22.5	0.42 ± 0.04	-12.93 ± 1.02

Values are presented as Mean ± SEM, $n = 4$.

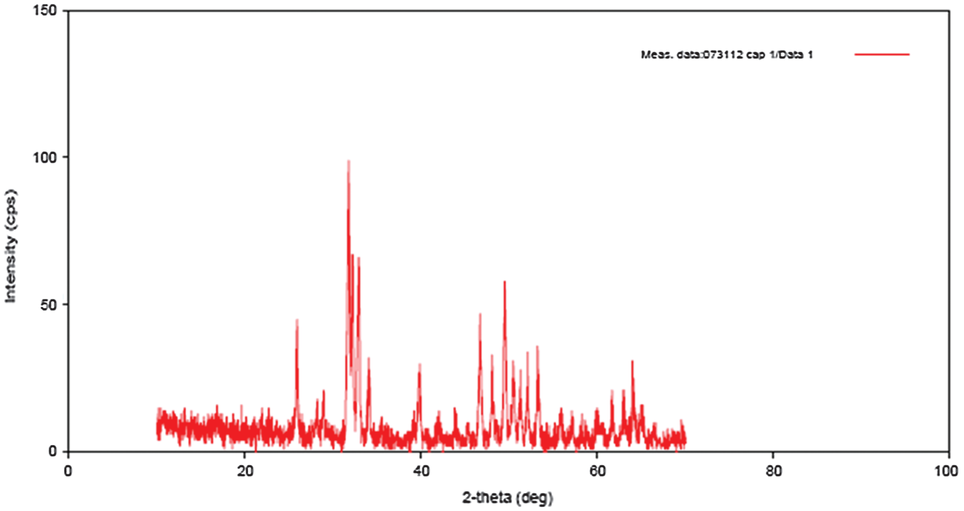

Fig. 1. X-Ray diffraction spectrum of CAP nanoparticles.

trehalose and sucrose were used in this study.[21,39] The amount of disaccharides adsorbed onto the CAP nanoparticles increased linearly with increasing concentration of disaccharides (Fig. 3(a)). The adsorption of sucrose was lower than the other two sugars ($p < 0.001$) particularly at higher concentration. The difference in adsorption patterns of the sugars is due to the difference in packing arrangement of sugar molecules used in the study.[39] In case of trehalose and cellobiose, there are six hydroxyl groups and 2 methanolic (CH_2OH) groups, while in case of sucrose, there are five hydroxyl and 3 CH_2OH groups.

For ovalbumin adsorption, 5 mgl/ml sugar concentration was used for preparing sugar coated CAP nanoparticles. This particular concentration (5 mg/ml) of sugar was used because at higher sugar concentrations ($\geq$8 mg/ml), the protein adsorption was found to decrease (data not shown). The adsorption of ovalbumin increased in a linear fashion till 0.2 mg/ml after which the adsorption curve reached a plateau (Fig. 3(b)). The plateau in the protein adsorption indicates saturation of adsorption sites on sugar coated CAP particles. The adsorption of protein onto the sugar coated CAP core is attributed to the formation of protein–sugar hydrogen bonds.[25] Ovalbumin showed significantly higher ($p < 0.05$) adsorption to cellobiose/trehalose coated CAP nanoparticles compared to sucrose coated CAP nanoparticles at protein concentration >1 mg/ml. Similarly Patil et al.,[39] found highest adsorption of hemoglobin on cellobiose adsorbed hydroxyapatite particles compared to sucrose adsorbed particles. Since the antigen is adsorbed rather than encapsulated in the matrix, the protein can retain its mobility and is accessible for interaction with immune cells and antibodies.

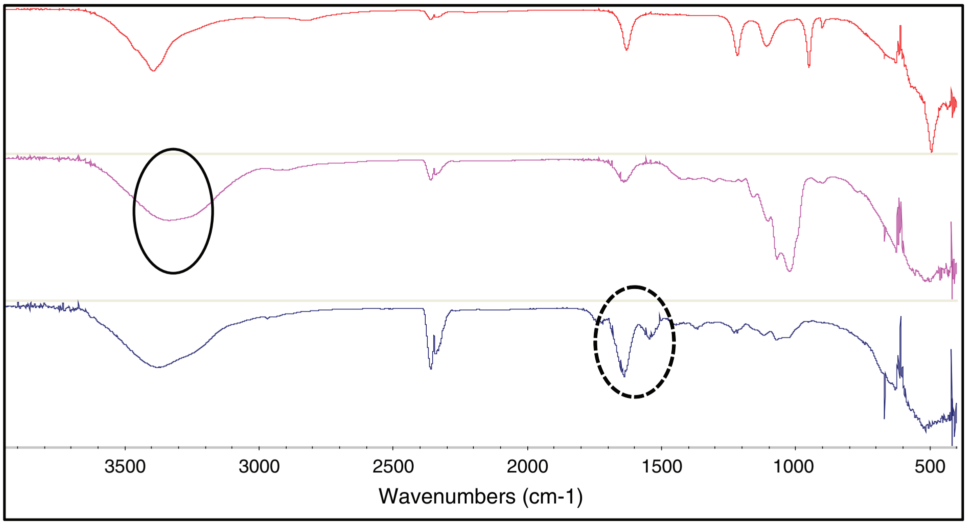

Fig. 2. ATR-FTIR spectra of CAP (a), CAP-cellobiose (b), and CAP-cellobiose-ovalbumin (c) nanoparticles. The thick circle represents the broadened hydroxyl peak indicating adsorption of cellobiose on CAP core. The dotted circle represents amide I and amide II peaks after adsorption of ovalbumin on particles.

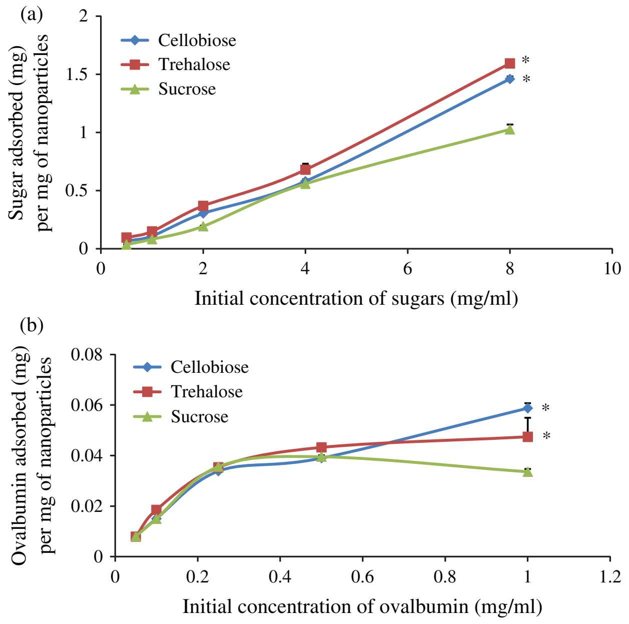

Fig. 3. Adsorption of three different sugars (trehalose, cellobiose and sucrose) onto CAP nanoparticles (a). The plot is represented as amount of sugars adsorbed (mg/mg of particles) versus concentration of sugars (mg/ml) used in the experiment. Adsorption of ovalbumin on sugar adsorbed CAP nanoparticles (b). The plot is presented as amount of ovalbumin adsorbed (mg/mg of particles) versus concentration of ovalbumin (mg/ml) used in the experiment. The values are presented as mean ± SEM, $n = 4$. (∗) represents that the values are significantly ($p < 0.001$) different compared to sucrose.

After adsorption of ovalbumin, the size of particles coated with cellobisoe, trehalsoe, and sucrose increased to 350 ± 22.5 nm, 450 ± 25 nm, and 500 ± 55 nm respectively. The increase in particle size after ovalbumin adsorption can be attributed to multiple layers of protein being adsorbed onto the particles. The differences in sugar-protein interactions can also contribute to the particle size of the nanoparticles. However, further studies are required to understand the mechanism of adsorption. Since the nanoparticles adsorbed with cellobiose were smaller in size compared to trehalose/sucrose adsorbed nanoparticles, cellobiose was used for all further experiments. The significant reduction in zeta potential of nanoparticles (−12.93 ± 1.02 mV) suggests the adsorption of ovalbumin, which has an isoelectric point of 4.9 and possesses net negative charge at physiological pH 7.4.[40] The ovalbumin adsorption on the nanoparticles was further confirmed using FTIR which showed peaks at 1654 cm^{-1} and 1550 cm^{-1} for amide I (═CO) and amide II (NH) respectively.[41]

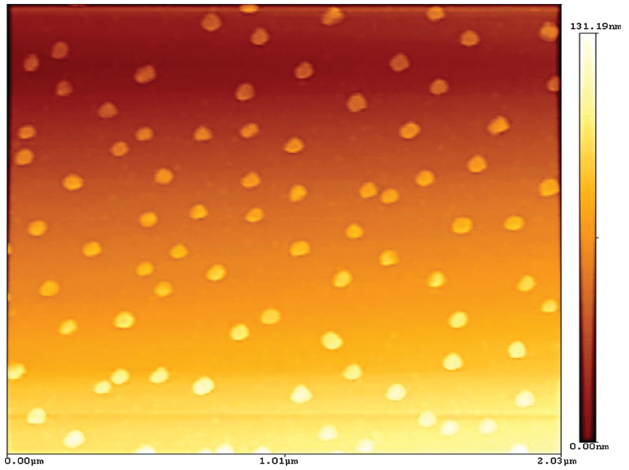

Fig. 4. AFM image of CAP-cellobiose-ovalbumin nanoparticles. AFM image was taken in tapping mode and represents two dimensional image of nanoparticles (2 μm × 2 μm scans).

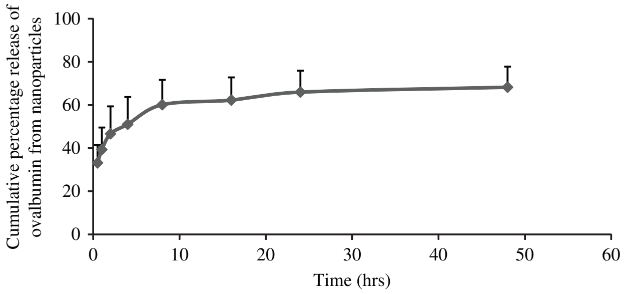

Fig. 5. *In-vitro* release study of ovalbumin from nanoparticles in phosphate buffer (pH 7.4) at 37 °C. Each data point is represented as mean ± SEM, $n = 3$–4.

Figure 4 shows the AFM image of CAP-cellobiose-ovalbumin nanoparticles. The nanoparticles were uniformly dispersed and spherical in shape with an average size of 129 ± 15 nm. The size of particles was smaller than found by DLS. The difference is attributed to the different experimental methods and mechanisms involved in preparing and analyzing the samples. DLS measures the hydrodynamic radius in liquid dispersion while the size measured by AFM is in the dry state. The immobilization affects the size and shape of particles when measured by AFM.[42] In a recent study comparing the two methods, DLS values for polystyrene nanoparticles were higher than AFM values.[43]

Adsorption efficiency of ovalbumin on cellobiose coated CAP nanoparticles was found to be 25 ± 3% and is consistent with previous reports.[25] The loading of ovalbumin

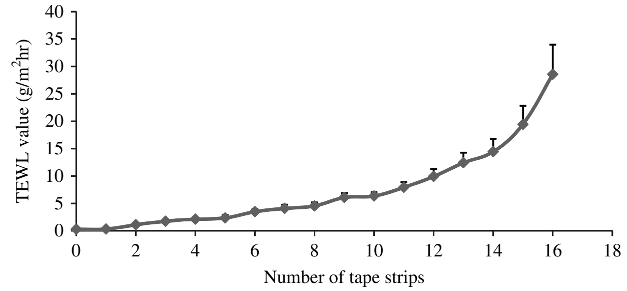

Fig. 6. Increase in TEWL values after sequential removal of the stratum corneum layers from mice abdominal skin using scotch tape. The values are presented as mean ± SEM, $n = 6$.

in CAP nanoparticles was 50 ± 5 mg/g. *In-vitro* release of nanoparticles in phosphate buffer (pH 7.4) at 37 °C showed a biphasic release (Fig. 5). Around 30% burst release was observed followed by a sustained release for 48 hours. The release profile is consistent with a recent study which reported sustained release of albumin from CAP nanoparticles for over 30 hrs.[25] The burst release of antigen helps to prime the immune system, while sustained release provides prolonged exposure of antigen to the antigen presenting cells for enhanced memory response.[44] Further optimization of particles by altering the Ca/P ratio and/or protein loading can be used to modify the release kinetics.

3.2. *In-Vivo* Immune Response

The *in-vivo* immunization studies were carried out in Balb/C mice to investigate the feasibility of transcutaneous

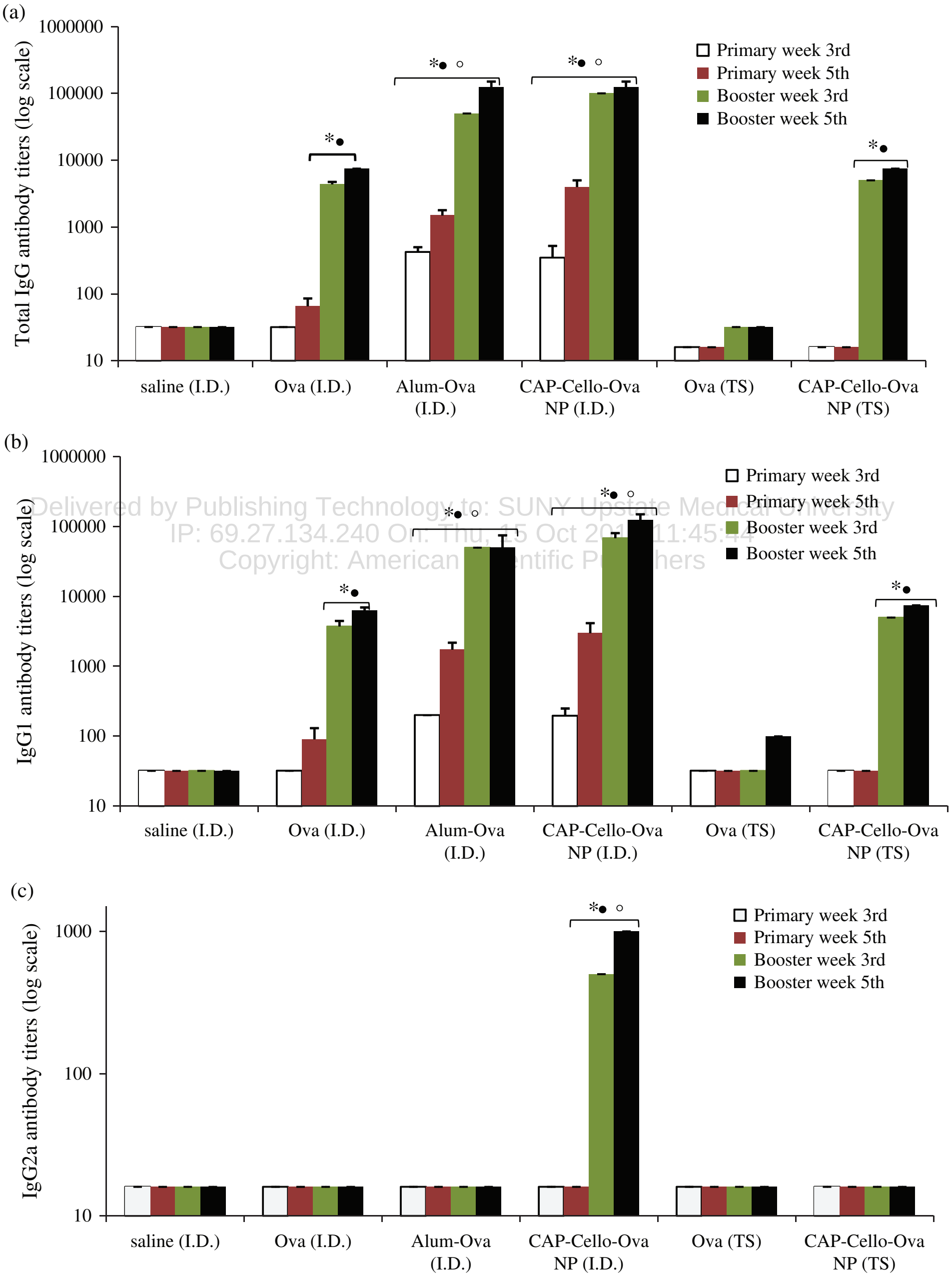

Fig. 7. Total IgG (a), IgG1 (b), and IgG2a (c) antibody titers after *in-vivo* immunization studies in Balb/C mice. The titers were analyzed using ELISA at 3rd and 5th weeks after primary as well as after booster doses. The y axis shows the antibody titers in log scale. The maximum dilution at which the absorbance value was still higher [Average + 2(SD)] than saline, was considered as the titer value. The results are presented as Mean ± SEM, $n = 3$–4 mice. (∗), (●) and (○) represent that the values are significantly different ($p < 0.01$) compared to saline, ovalbumin (TS) and ovalbumin (ID) respectively in the corresponding weeks. TS and ID represent tape stripping and intradermal injection respectively.

vaccine delivery using CAP nanoparticles on tape stripped skin. The major challenge in transcutaneous vaccine delivery is to overcome the stratum corneum (SC) barrier. In this regard, several physical penetration enhancement approaches including simple hydration, scarification, abrasion, and tape stripping have been explored.[3,4,7] Since our goal was to test the feasibility of using CAP nanoparticles for transcutaneous vaccine delivery, we used tape-stripping procedure to remove the SC layer. Figure 6 shows the increase in TEWL values after sequential removal of SC layer from mice abdominal skin using scotch tape. There was a significant increase in TEWL after 15–16 tapes and the TEWL value reached about 20–25 g/m^{2}hr. This value is representative of complete removal of stratum corneum.[45] Seo et al.,[46] showed that topical application of tumor associated antigenic peptides onto tape stripped mice skin induced significant tumor-specific cytotoxic T lymphocytes (CTLs) and protected mice against subsequent challenge with corresponding tumor cells. Tape stripping can increase the expression of stimulatory molecules (CD86, CD54 and MHC class II) and antigen-presenting capacity of epidermal dendritic cells.[47] Also, tapestripping has been shown to result in the production and release of various cytokines including IL-1α, IL-β, and TNF-α, and IFN-γ which modulate immune response.[48,49]

Figure 7 shows the total IgG, IgG1 and IgG2a anti-ovalbumin antibody titers at 3rd and 5th week after primary and booster immunization. To test the ability of the CAP nanoparticles as a vaccine adjuvant intradermal injection was used. There was a time dependent increase in antibody titers after immunization in all treatment groups. Intradermal administration of CAP-cellobiose-ovalbumin nanoparticles showed significantly ($p < 0.01$) higher antibody titers (Total IgG, IgG1) compared to ovalbumin alone. The titers were comparable to the positive control (alum-ovalbumin). CAP-cellobiose nanoparticles were also studied to investigate if these particles by themselves induce any non-specific immune response. However, no significant antibody response was observed compared to saline and ovalbumin (data not shown). IgG2a response was also found to be higher in mice immunized intradermally with particles compared to ovalbumin and alum-ovalbumin (Fig. 7(c)). The effect of physical mixture of ovalbumin and CAP cellobiose nanoparticles was also investigated. Although the physical mixture showed significantly higher ($p < 0.01$) total IgG and IgG1 titers compared to ovalbumin, the response was less compared to ovalbumin adsorbed particles (data not shown). Moreover, the physical mixture did not produce any IgG2a response (data not shown). The sugar serves as a ligand and binds to the sugar-like mannose receptors on APCs such as macrophages and dendritic cells.[25] Therefore, the enhanced immunological response of ovalbumin adsorbed sugar coated CAP nanoparticles can be attributed to the better presentation and uptake of the nanoparticles by the APCs. Further the sustained release of antigen enhances the antigen presentation to immune cells and hence improves the immune response.

Topical application of ovalbumin coated CAP nanoparticles on tape stripped mice skin significantly enhanced the antibody titers (Total IgG and IgG1) compared to ovalbumin (Figs. 7(a and b)). Not unexpectedly the magnitude of antibody response was lower compared to intradermal injection. Topical application of particles did not show any IgG2a response (Fig. 7(c)). This could be attributed to the relatively shorter contact time of the formulation with tape stripped skin (4 hrs) compared to the intradermal injection. Further intradermal injection can deliver a larger number of CAP nanoparticles to the dermis layer. The stimulation of IgG2a antibodies requires higher doses compared to IgG1 antibodies.[50] The optimization of the application time and skin penetration enhancement techniques such as microneedles or jet injections, can improve the immune response. Nevertheless, the results suggest the feasibility of topical vaccine delivery using CAP nanoparticles as a vaccine adjuvant/delivery vehicle.

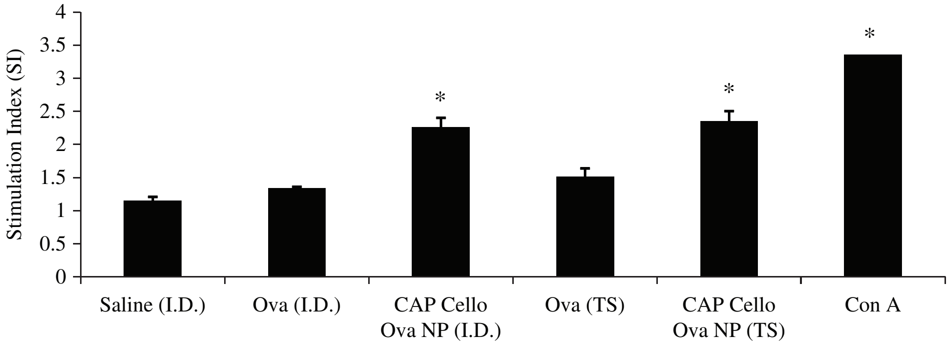

Fig. 8. The splenocytes of preimmunized mice were stimulated with RPMI alone, ovalbumin and Con A mitogen for 72 h at 37 °C in a CO_2 incubator and the proliferation was measured using MTT assay. Stimulation index (SI) was calculated by dividing the absorbance value of ovalbumin or Con A treatment group with that of RPMI treated group. The values are presented as mean ± SEM, $n = 3$–4 mice. (∗) represents that the values are significantly different ($p < 0.01$) compared to saline and ovalbumin. TS and ID represent tape stripping and intradermal injection respectively.

The *in-vitro* proliferation of splenocytes was investigated after re-stimulation with the same antigen. It represents the ability of memory T cells to recognize the antigen upon re-stimulation. Generation of a long term memory response is one of the key features of an effective vaccine delivery system.[51] The results of *in-vitro* splenocyte proliferation assay is shown in Figure 8. The stimulation index (SI) was calculated by dividing the absorbance values obtained after treatment with ovalbumin/ConA mitogen with values obtained after treatment with RPMI media. The increased proliferation of splenocytes in mice treated with CAP-cellobiose-ovalbumin nanoparticles by intradermal and TCI suggest the ability of CAP nanoparticles to enhance the antigen presentation to the T cells compared to ovalbumin alone. Both the routes showed comparable stimulation index suggesting the ability of ovalbumin adsorbed CAP nanoparticles to induce long lasting immune response. Further mechanistic studies including LC activation and lymph node migration assay, dendritic cell maturation, and antigen presentation to T cells are required to understand the vaccine adjuvant effect of calcium phosphate nanoparticles through skin.

4. CONCLUSIONS

Results from this study demonstrate the feasibility of transcutaneous vaccine delivery using CAP nanoparticles. Ovalbumin loaded CAP nanoparticles enhanced the antibody titers as well as the *in-vitro* splenocyte proliferation compared to ovalbumin alone. Optimization of nanoparticles including adsorption efficiency and release kinetics could improve the adjuvant properties of CAP nanoparticles. Further enhancing the penetration of particles into deeper skin layers using other physical penetration techniques such as microneedles can improve the immune response through skin.

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